

Effectivity

The following CT Equipment is affected:

2266832	LightSpeed Plus Console Assembly
2266832-2	LightSpeed Plus Console Assembly w/D3D

Purpose

The purpose of this FMI is to upgrade the Image Generator (IG) circuit board found in the LightSpeed Plus Console to the new Pegasus Image Generator. The Pegasus board replaces both the current IG board and the "Mercury" scan data corrections (SDC) board.

Prerequisite FMIs

FMI25275: SCSI Upgrade

Related FMI

FMI25277: H2 M2.1 Software Upgrade

Furnished Materials

FMI Kit Part Number 2296264 includes the items listed below.

Qty	Description	Part #
1	FMI Document	2291959-100
1	Pegasus Image Generator Board	2216467
5	VME Backplane Jumpers	46-221852P2
2	Knurled Thumbscrews	2280022
1	Medium Metal Filler Panel	2280020
2	Metal Spacer Bars	2287121
4	Screws (for Spacer Bars)	46-328421P5
1	Ty-Rap Mount	46-208747P2
AR	Ty-Raps	46-208758P1

Required Tools and Test Equipment

Qty	Description	Part #
1	Phillips Head Screwdriver	
1	Standard (Flat-blade) Screwdriver	
1	Small Holding Screwdriver (flat-blade)	e.g., VACO K-23
1	ESD Wrist Strap	
1	Metric Hex Wrench set (inc. 2mm & 4mm)	
1	Needle Nose Pliers (for jumpers)	
1	Diagonal cutters (for ty-raps)	
1	Flashlight or Trouble-light	
1	5mm Hex Socket, extension and drive	

Removed Parts & Disposal

The following parts will be removed from the console during this procedure, and will not be reinstalled:

Qty	Description	Part #
1	Rhapsode Image Generator board	2118899-2
1	Mercury SDC board	2185174
2	Circuit Board Retaining Brackets	2264874
1	Filler Panel (small)	2214515-2
	Assorted screws	

U.S. ONLY

- 1.) Place the IG and SDC boards into separate static bags (for the old IG board, reuse the bag used to ship the Pegasus board).
- 2.) Seal the bags and reuse the pegasus box for shipment.
- 3.) Label and mail to:
Repair Development Center
Attn: Paul Kolvenbach
430 W. Grange Ave.
Milwaukee, WI 53207
- 4.) Destroy and/or dispose of all obsolete hardware, software and documentation according to GE policy.

INTERNATIONAL SITES (OUTSIDE THE U.S.)

Destroy and/or dispose of all obsolete hardware, software and documentation according to GE policy.

People Power

Allow 2 hours for hardware installation and an additional 2 hours to load software.

Procedural Overview

Please remember to **save your system's state**. The following procedure installs new software on the CT scanner.

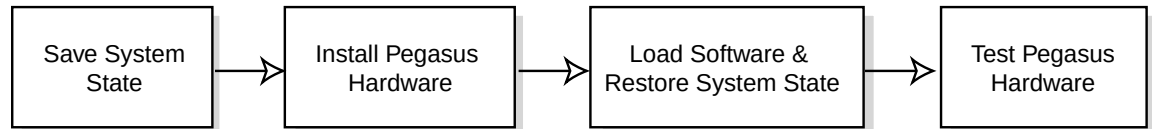


Figure 1 Procedural Overview

- Note:
- Perform SCSI Upgrade FMI25275 (if applicable), before proceeding.
 - DO NOT load software (i.e., FMI25277) until after all hardware changes are complete.

Install Pegasus Hardware

1 Power Shutdown & Cover Removal

Purpose: The purpose of this section is to 1) Ensure that no circuits are live within the Console, and 2) Remove the console covers to gain access to components.

- 1.) Bring software down.
- 2.) Shutdown system, following normal shutdown procedures.
- 3.) Tag and lockout the main system power, supplied to the PDU (i.e., main disconnect).



WARNING



POTENTIAL FOR PERSONAL INJURY OR DEATH BY ELECTRIC SHOCK. TO AVOID RISK OF ELECTRIC SHOCK, FOLLOW PROPER LOCKOUT/TAGOUT PROCEDURES.

- 4.) Disconnect J19, J20 and J52 from the rear of the console. Remove the back panel by removing the ten (10) screws that hold it in place.

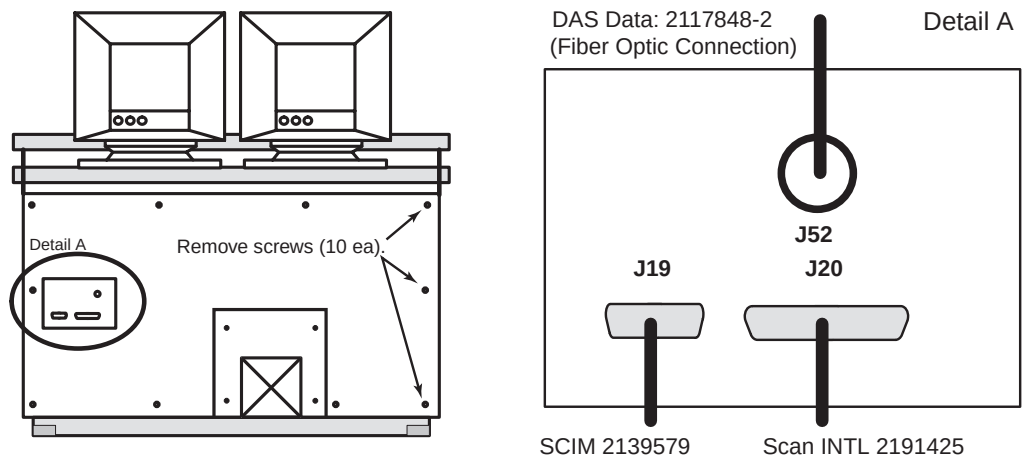


Figure 2 Rear of Console, showing bulkhead and screw locations

- 5.) Using a 4mm hex wrench, remove the console's front cover.

- 6.) Remove the front EMC cover from the "ICEbox".

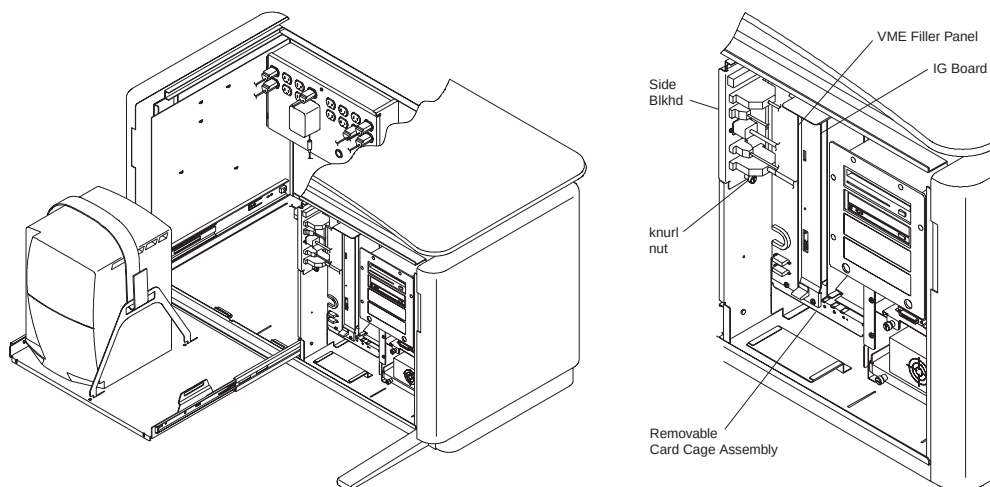


Figure 3 Console, with front cover and EMC cover removed / Detail of "ICEbox"

2 Parts Removal

Purpose: Because the Scan Data Corrections (Mercury) board is eliminated via this procedure, additional jumpers must be added to the VME backplane. This means that all of the boards must be removed from the card cage assembly (Figure 4), so that the backplane itself may be removed for the installation of said jumpers. This section steps through the removal of parts to gain access to the VME backplane. Removal of the VME card cage is not required.

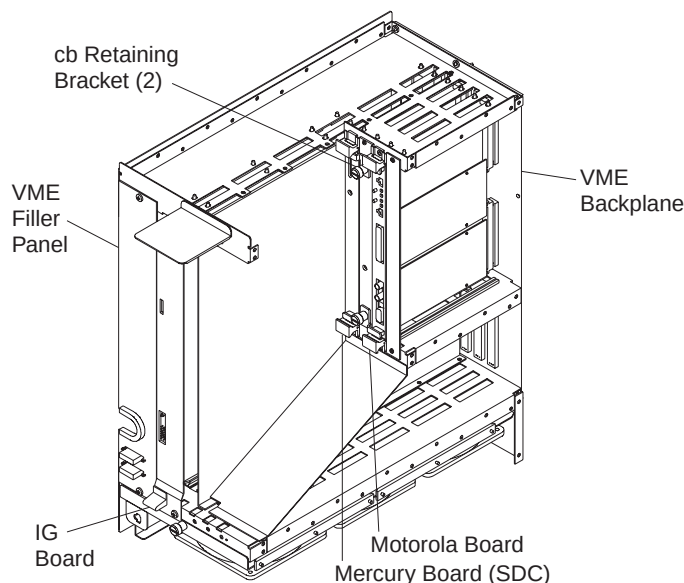


Figure 4 Removable Card Cage Assembly

- 1.) Disconnect cables from the Motorola board.

- 2.) Remove the side bulkhead assembly (see Figure 5) by loosening the knurl nut. Position the bulkhead so that it is out of the way of the card cage assembly (use care handling the ribbon cable).

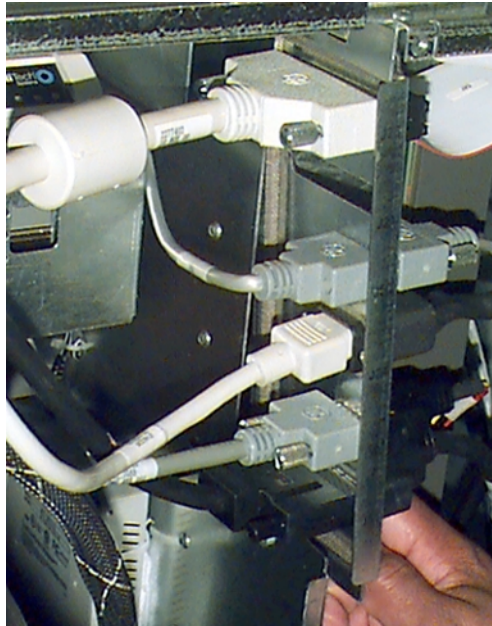


Figure 5 Side Bulkhead Removal

- 3.) Disconnect J30 & J31 (see Figure 6) connectors from the VME filler panel (located at the left of the card cage). Remove the panel, retaining the two (2) screws for reinstallation.



Figure 6 ICE Box Connectors J30 and J31

- 4.) Remove the four (4) 2mm allen head screws that hold the Image Generator (IG) circuit board in place. Retain these screws and the two (2) nylon washers for later reuse.



NOTICE



Wear a grounded ESD wrist strap. Place removed electronic parts on an anti-static surface.

- 5.) Remove the IG board and place it into a static bag.

- 6.) Reach into the left side of the chassis (card cage) and disconnect the J15, J6 and J5 power cables connected to the VME Backplane.
- 7.) Loosen the thumbscrews, and remove the two (2) circuit board retaining brackets (Figure 7) from within the card cage (see Figure 4 for location). These brackets will not be reused.



Figure 7 Circuit Board Retaining Bracket

- 8.) Remove the Mercury SDC circuit board and place into a static bag. Retain the screws for later reuse with a new filler panel.
- 9.) Remove the Motorola RIP circuit board and place into a static bag. Retain the screws for reinstallation.
- 10.) Remove the filler panel located to the left of the Motorola RIP circuit board slot. Retain the two (2) screws for later reuse.
- 11.) Remove the rear cover of the "ICEbox."
- 12.) Locate the VME backplane at the rear of the console. Remove and retain the ten (10) phillips screws and washers attaching the VME backplane to the card cage chassis. Use caution when removing the last screw, so as to not let the backplane fall.

Power Lead Connections

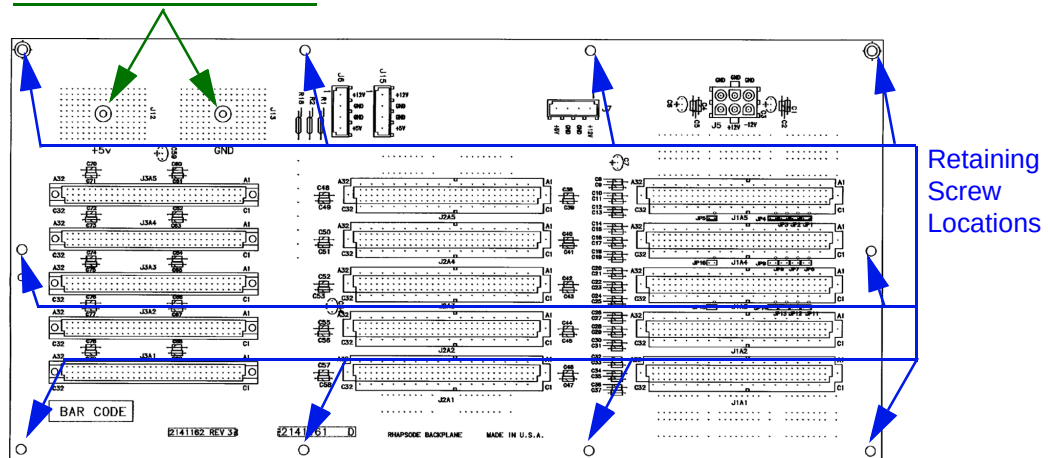


Figure 8 VME Backplane Physical Connections



NOTICE
Potential for
Equipment
Damage

Handle the VME backplane carefully. It's recommended that you **DO NOT** remove power leads from VME backplane. The power connections on this assembly have been torqued to 45 inch-pounds.

3 Parts Installation

Purpose: This section steps the user through the installation of jumpers on the VME backplane (required due to the removal of the Mercury SDC board), the installation of the new Pegasus Image Generation board, the reinstallation of the Motorola (RIP) board and various connectors, and the reassembly of the Console.

- 1.) Verify that the new retaining “thumb” screws (Figure 9) fit properly. New retaining screws are used to secure the Motorola board in its slot.



Figure 9 Retaining Screws (both old and new)

Carefully slide the Motorola board into its slot (Slot 1, see Figure 10) without the backplane attached. Insert the new retaining screws and lock the board into place.

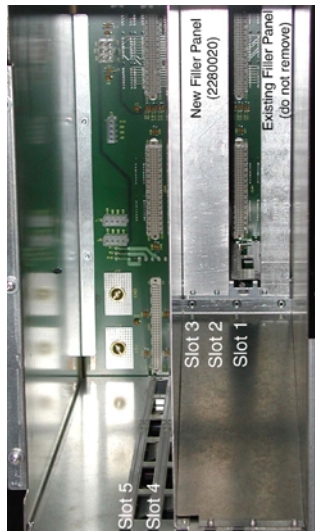


Figure 10 VME Chassis Slot Identification



NOTICE
Do not force
thumbscrews

Do not force the thumbscrews. In the unlikely event that the new thumbscrews will not go in, then the VME card retaining brackets will need to be adjusted. The bottom card retaining bracket is accessible from the rear of the console, with the backplane removed (see Figure 11). Access to the top bracket can be gained only by removing the VME card cage from the console.

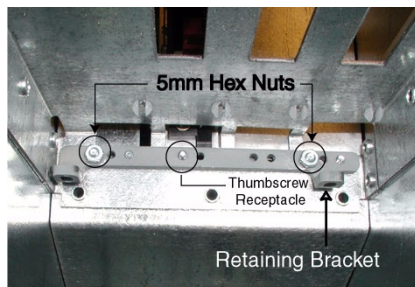


Figure 11 Bottom VME Card Retaining Bracket (rear view, backplane removed)

- 2.) Get five (5) jumpers from the kit. Position the VME backplane circuit board to gain access to the connector side. Install the five (5) jumpers at JP6 through JP10 (see Figure 12), with the notched end UP. You may need to use needle nose pliers to insert the jumpers.

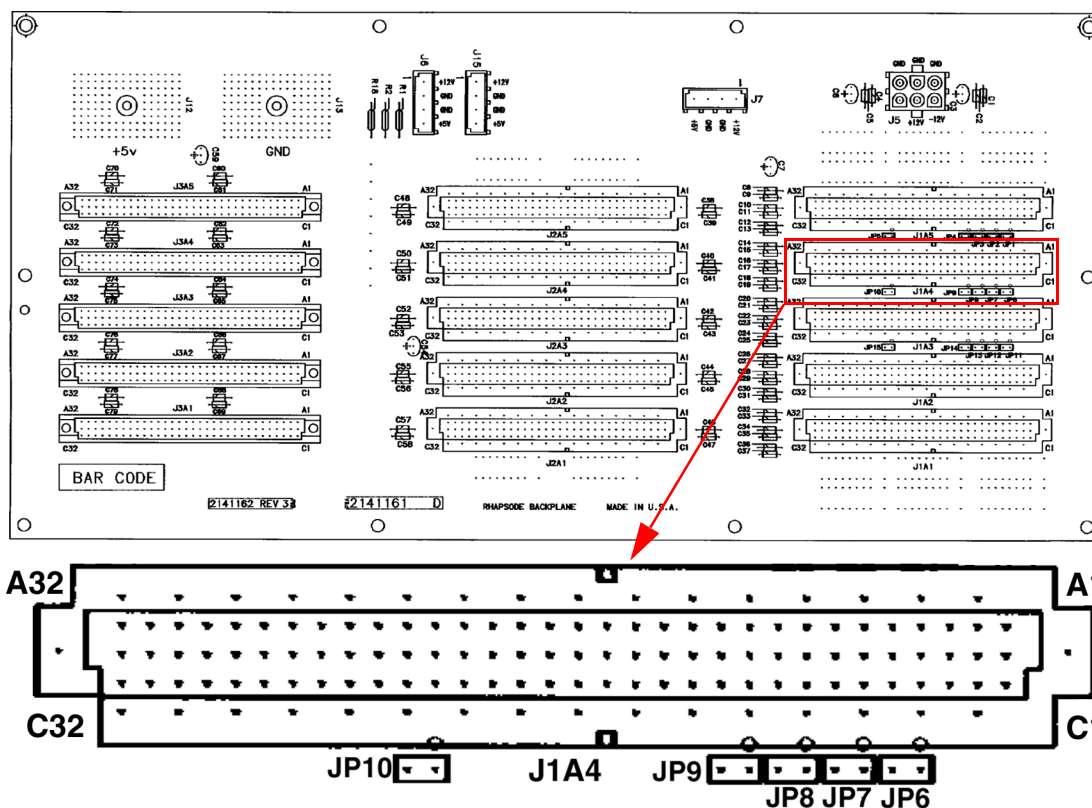


Figure 12 VME Backplane (top), showing Jumper locations (bottom)

- 3.) Reach into the card cage chassis and reconnect the J15 (to J30 on VME filler panel), J6 (to J31 on VME filler panel) and J5 power cables to the VME backplane. Ty-rap the cables to the side of the card cage.
- 4.) Reinstall the ten (10) phillips screws and washers to attach the VME backplane to the chassis.
- 5.) Reinstall the VME filler panel (with J30 & J31), using two (2) screws, at the left end of the chassis.
- 6.) Procure the new filler panel from the kit. Locate Slots 3 and 2 in the card cage. Install the panel IN the chassis using the two (2) screws previously removed (from the old filler panel). Orient the panel with the narrow edge at the left, and set the panel with its left edge flush to the chassis wall. Mount the plate using the top left and bottom right holes (see Figure 13).



Figure 13 Filler panel orientation and screw locations (not to scale)

- 7.) Procure from the kit, the two (2) metal spacer bars and their four (4) screws.
- 8.) Install a metal spacer bar at the top and bottom of the card cage assembly face (*not* inside the assembly). Orient each spacer so it can be attached via two (2) screws (46-328421P4) positioned at Slots 3 and 1.

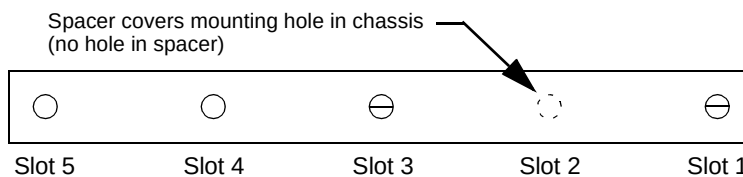


Figure 14 Orientation of Spacer Bars

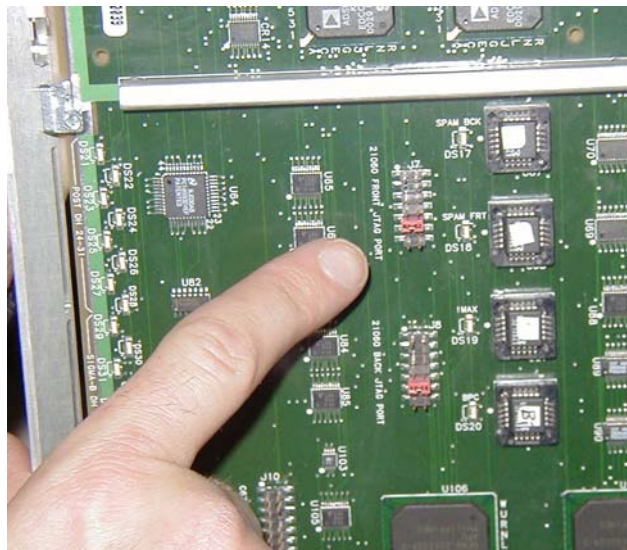


NOTICE



Wear a grounded ESD wrist strap.

- 9.) Remove the Pegasus Image Generator circuit board from its static bag, and verify that it has only two (2) jumpers installed. As shown in Figure 15:
 - J7: positions 9 and 10
 - J8: positions 9 and 10



Jumpers

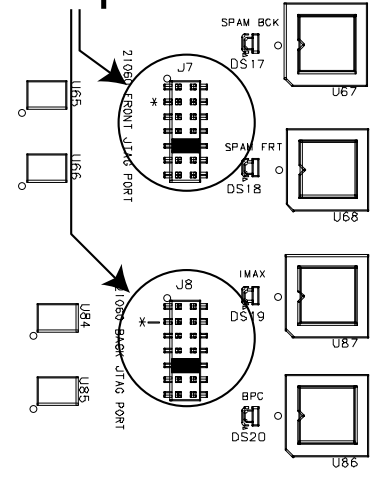


Figure 15 Pegasus IG Jumpers

- 10.) Inspect the connectors on the Pegasus IG board, to verify that they are free from foreign objects and that no pins are bent.

- 11.) Install the four (4) allen head screws (2142006) and two (2) nylon flat washers (2104490) on the Pegasus Image Generator circuit board faceplate. The washers should be used for the mounting holes on the "tabs" at the left of the faceplate.
- 12.) Place the Pegasus Image Generator circuit board into the card cage and tighten the four (4) allen head screws to properly secure the board.
- 13.) Reinstall the side bulkhead assembly. Route the ribbon cable above as usual.
- 14.) Procure the two (2) knurled thumb screws from the kit, and install them into the faceplate of the Motorola RIP board.
- 15.) Reconnect the cables to the Motorola RIP board, as shown in Figure 16. Then reinstall the board, using the knurled thumbscrews.

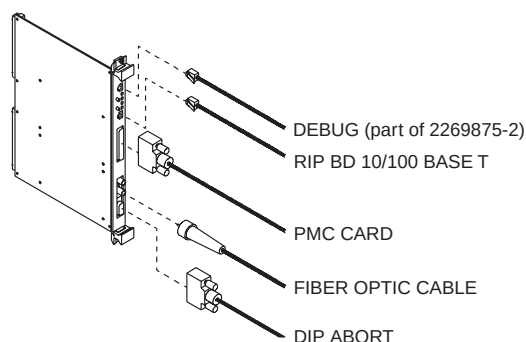


Figure 16 Motorola Board - Connections

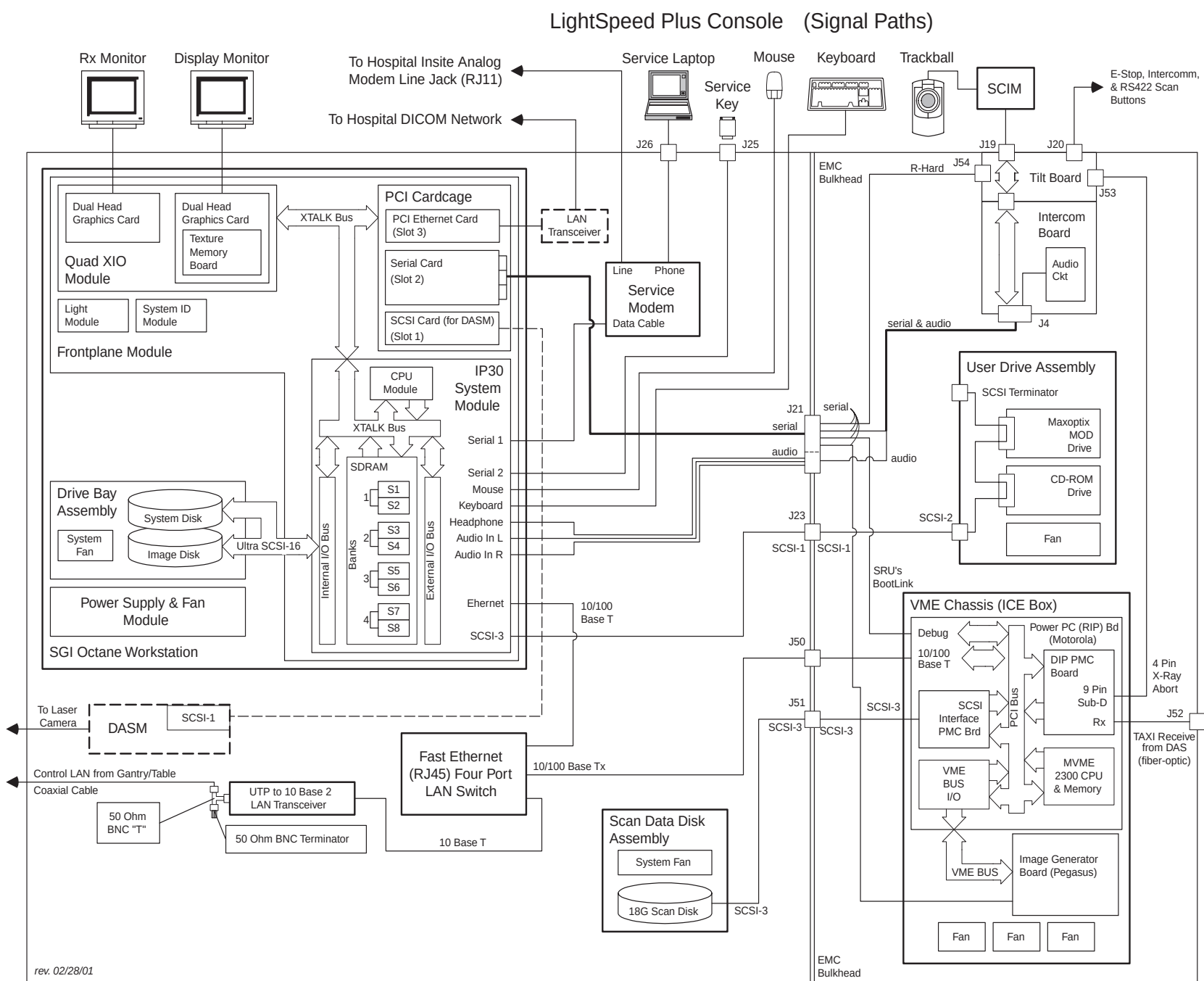
- 16.) Connect the Pegasus Cable to the Pegasus Image Generator circuit board faceplate location labeled "Serial Port."
- 17.) Reconnect J31 and J30 to the VME filler plate.
- 18.) Procure the Ty-Rap Mount and Ty-Rap Strap from the kit. Position the Ty-Rap Strap through the Ty-Rap Mount holes. Place the Ty-Rap Mount on the horizontal panel directly below (under) the User Drive Tray. Secure the Ty-Rap Strap around the Fiber Optic Cable. This will position the cable away from the Intercom Board.
- 19.) Replace the "ICEbox" front and rear EMC covers.
- 20.) Replace the console's front and rear covers.
- 21.) Reattach connectors J19, J20 and J52 to the rear bulkhead.



Figure 17 Finished Connections in the ICE

- 22.) Reapply power.

DIRECTION 2291959-100, REVISION 2



Load System Software & Restore System State

Proceed to FMI25277 to load new system software. Software installation should only be performed if all console hardware changes are complete (i.e., FMI25275, if applicable, and this FMI).

Test Pegasus Hardware

Purpose: The Pegasus IG hardware should be tested prior to customer turnover, using the supplied diagnostic tools. Both Recon Data Path and Quick IG Diagnostics should pass one (1) iteration each without error. The following material describes how to run those diagnostics.

1 Diagnostic Overview

There are two (2) levels of Pegasus diagnostics:

- **Low level board level diagnostics** - run from the service browser with applications shutdown. Used to test the functionality of the Pegasus board specifically.
- **High level Recon Data Path tests** - run from the service desktop at applications level. Tests the ability of the Pegasus board to communicate and operate with the rest of the system.

It's recommended that low level diagnostics be executed first. Followed by high level diagnostics next.

1.1 Board Level Diagnostics

- 1.) With Applications shutdown, open a Unix shell. On the Service Desktop page, click UTILITIES (circular button at the top menu bar), then click Application Shutdown in the left menu column. Wait for the "Attention" window to disappear. Open a shell.

Within that shell, execute the command to start the service browser (See Figure 19):

service_browser ENTER

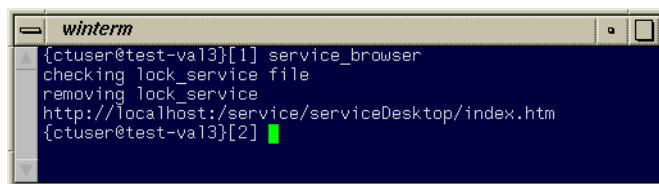


Figure 19 Initiating the Service browser

- 2.) With the service browser displayed, select the **Diagnostics** TAB (See Figure 20).

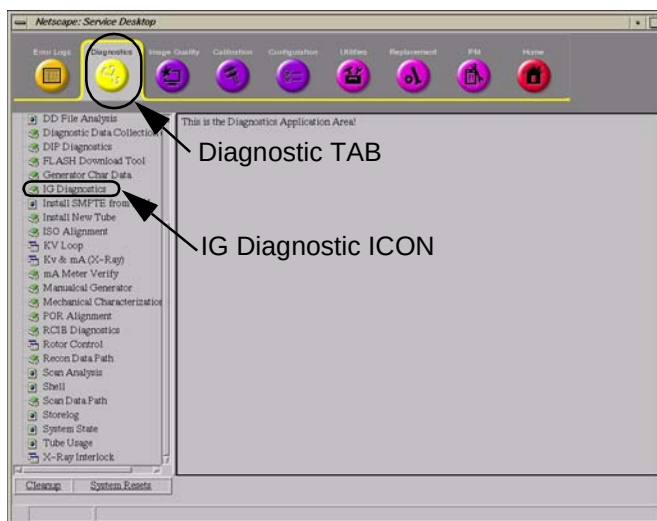


Figure 20 Service Browser - Diagnostic TAB

- 3.) Within the Diagnostic TAB, select the IG Diagnostic ICON in the file list menu on the left, by clicking on it. The IG Board level Diagnostic screen appears. (See Figure 21)

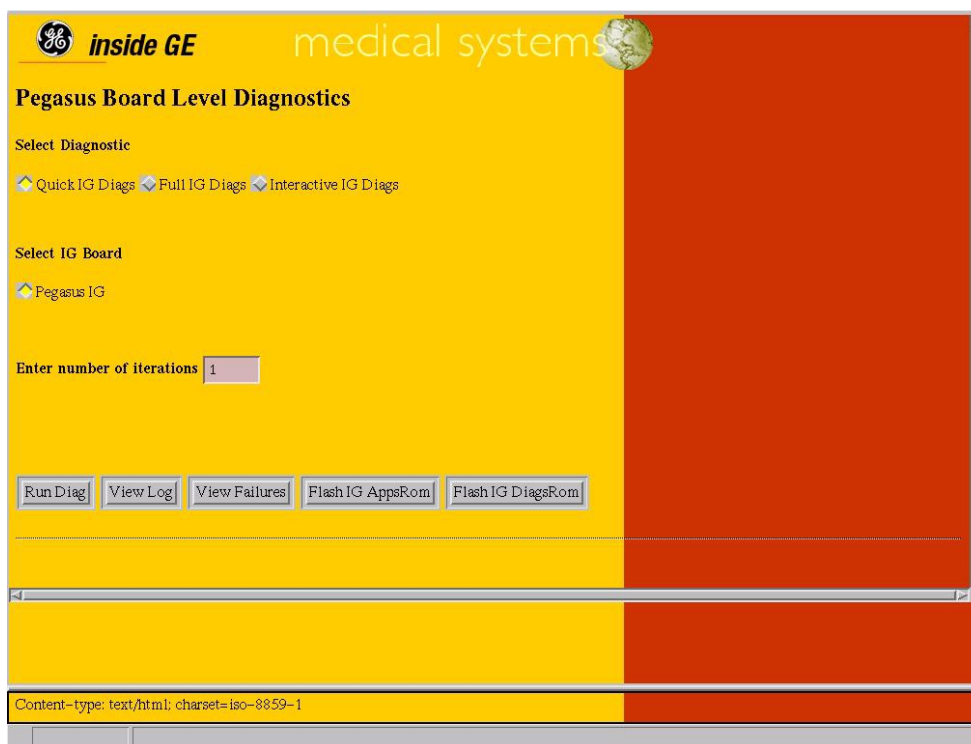


Figure 21 IG Board Level Diagnostic Screen

- 4.) Select either Quick IG Diags or Full IG Diags and Enter number of iterations.
- 5.) Select RUN DIAG. A shell window will open and the results of each test displayed on-screen. See Figure 23 and Figure 24.
- 6.) To exit testing, close the window. See Figure 26.

1.1.1 Run Diag

RUN DIAG is used to initiate the diagnostics chosen. Select RUN DIAG to begin test execution according to the parameters selected. First, select the diagnostic. Next, select the IG Board to be tested, Finally, enter the number of test iterations (1 is the default) desired.

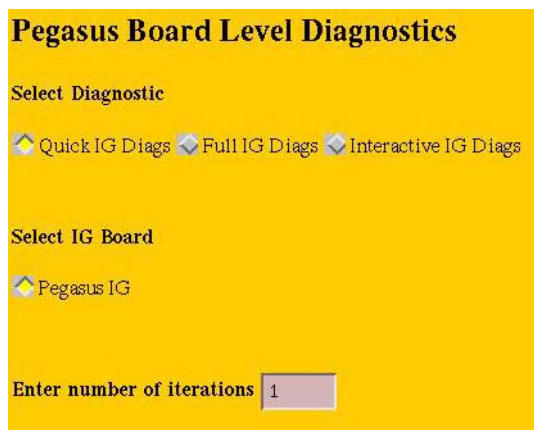


Figure 22 Board Level Diagnostic Parameters

A new window is displayed (spawned) with output from the test selected displayed.

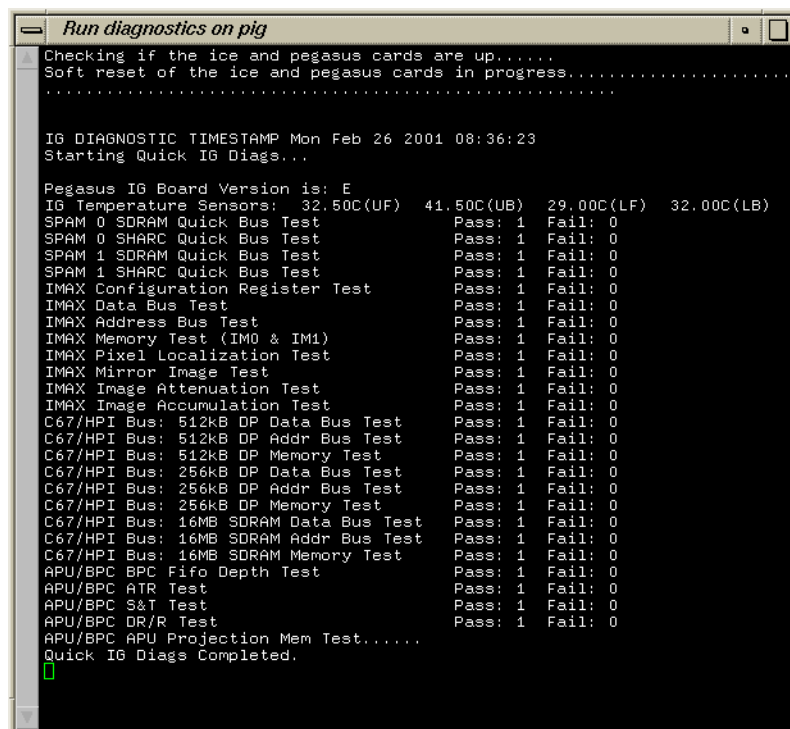
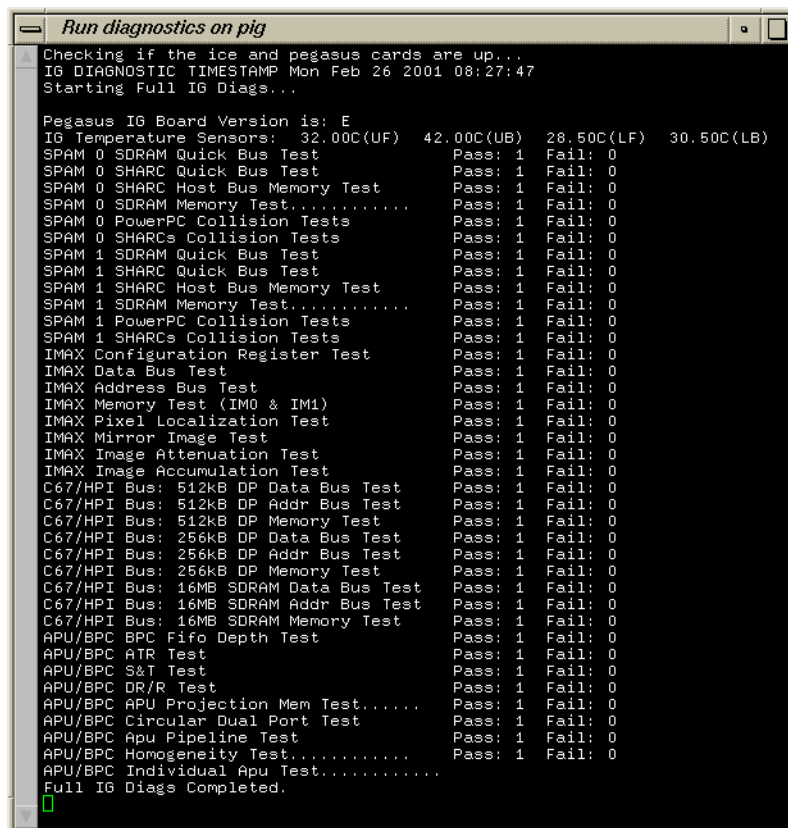


Figure 23 Quick Diags Screen



```
Run diagnostics on pig
Checking if the ice and pegasus cards are up...
IG DIAGNOSTIC TIMESTAMP Mon Feb 26 2001 08:27:47
Starting Full IG Diags...

Pegasus IG Board Version is: E
IG Temperature Sensors: 32.00C(UF) 42.00C(UB) 28.50C(LF) 30.50C(LB)
SPAM 0 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 0 SHARC Quick Bus Test Pass: 1 Fail: 0
SPAM 0 SHARC Host Bus Memory Test Pass: 1 Fail: 0
SPAM 0 SDRAM Memory Test..... Pass: 1 Fail: 0
SPAM 0 PowerPC Collision Tests Pass: 1 Fail: 0
SPAM 0 SHARCs Collision Tests Pass: 1 Fail: 0
SPAM 1 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SHARC Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SHARC Host Bus Memory Test Pass: 1 Fail: 0
SPAM 1 SDRAM Memory Test..... Pass: 1 Fail: 0
SPAM 1 PowerPC Collision Tests Pass: 1 Fail: 0
SPAM 1 SHARCs Collision Tests Pass: 1 Fail: 0
IMAX Configuration Register Test Pass: 1 Fail: 0
IMAX Data Bus Test Pass: 1 Fail: 0
IMAX Address Bus Test Pass: 1 Fail: 0
IMAX Memory Test (IM0 & IM1) Pass: 1 Fail: 0
IMAX Pixel Localization Test Pass: 1 Fail: 0
IMAX Mirror Image Test Pass: 1 Fail: 0
IMAX Image Attenuation Test Pass: 1 Fail: 0
IMAX Image Accumulation Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Memory Test Pass: 1 Fail: 0
APU/BPC BPC Fifo Depth Test Pass: 1 Fail: 0
APU/BPC ATR Test Pass: 1 Fail: 0
APU/BPC S&T Test Pass: 1 Fail: 0
APU/BPC DR/R Test Pass: 1 Fail: 0
APU/BPC APU Projection Mem Test..... Pass: 1 Fail: 0
APU/BPC Circular Dual Port Test Pass: 1 Fail: 0
APU/BPC APU Pipeline Test Pass: 1 Fail: 0
APU/BPC Homogeneity Test..... Pass: 1 Fail: 0
APU/BPC Individual APU Test..... Pass: 1 Fail: 0
Full IG Diags Completed.
```

Figure 24 Full Diags Screen

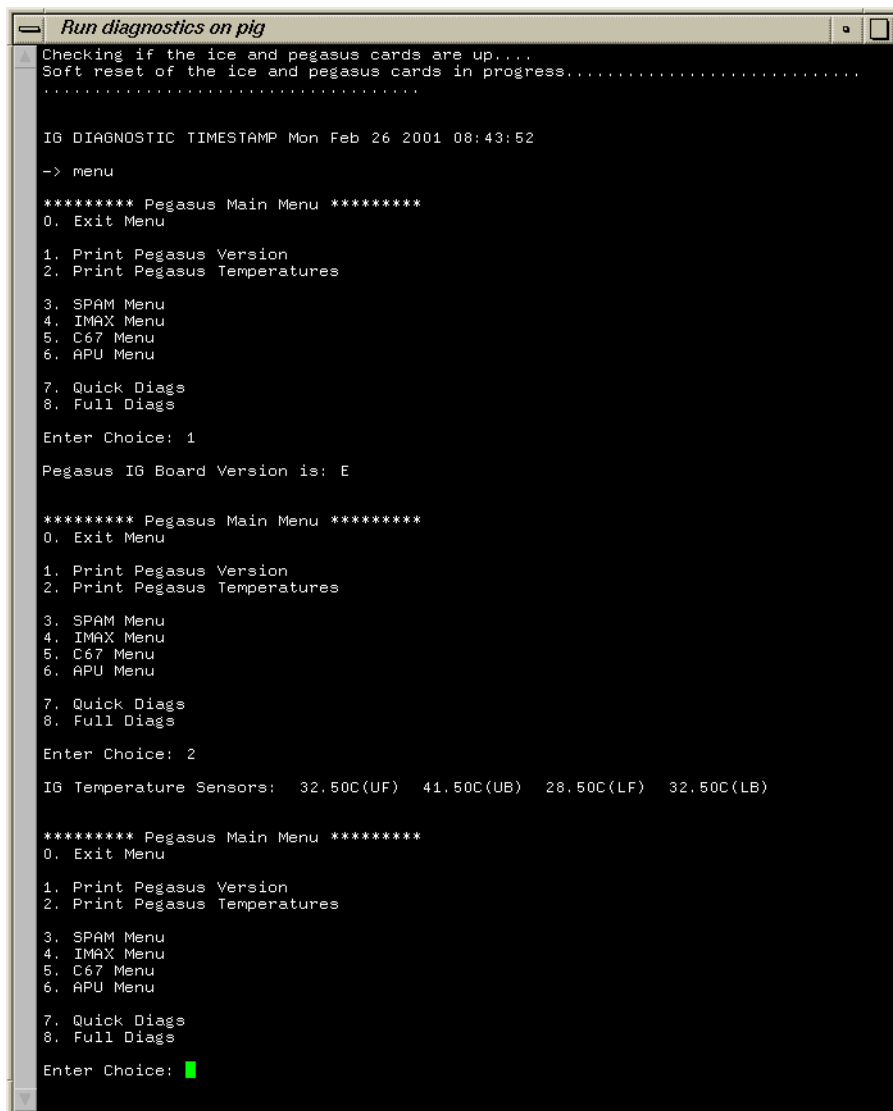


Figure 25 Interactive Diags Screen

Always close the test window after testing has completed. Double-click the square box in the upper left corner of the window with the minus sign.

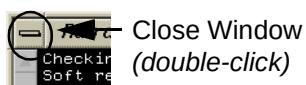


Figure 26 Close Window ICON

1.1.2 View Log

VIEW LOG displays the entire contents of the `pig.log`. Located in `/usr/g/service/log` directory.

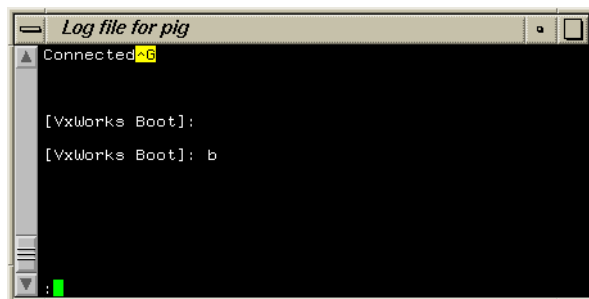


Figure 27 View Log Screen

1.1.3 View Failures

VIEW FAILURES displays all the IG failures recorded in the `pig.log`. Located in `/usr/g/service/log` directory.

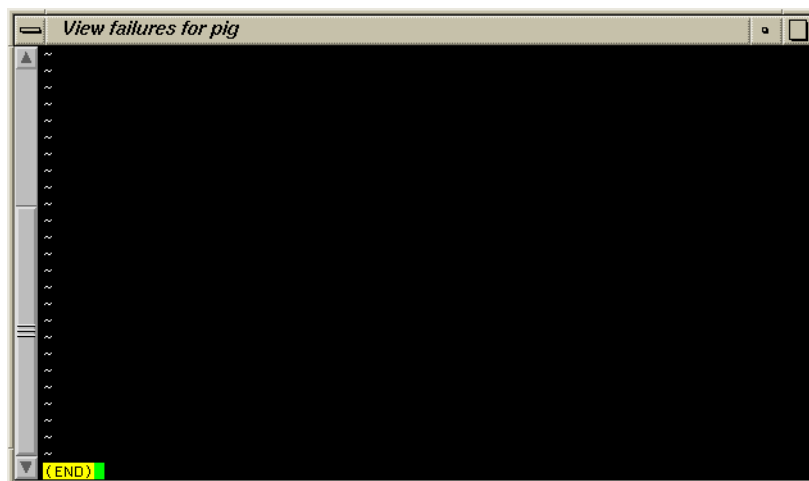


Figure 28 View Failures Screen

1.1.4 Flash IG AppsRom & Flash IG DiagsRom

Only use when instructed. For manufacturing use only.

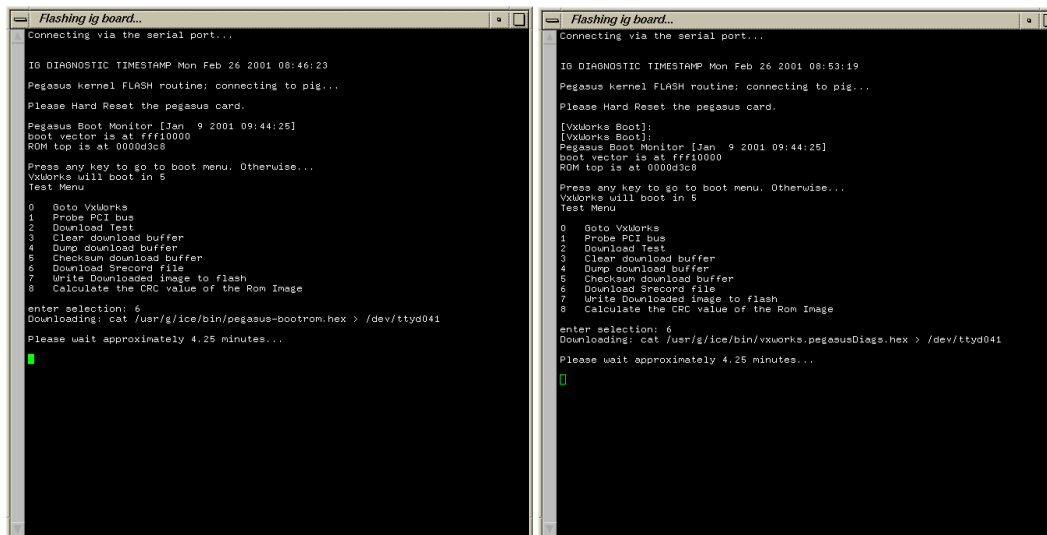


Figure 29 Flash IG AppsRom & DiagsRom Screen

1.2 Recon Data Path Test

CT Scanner applications must be up and running. If not, click on APPLICATION STARTUP, in the upper right corner of the screen. This takes about 2 minutes 40 seconds to come up. Applications are up when the a dialog window says that you need to run *fastcal*. Click OK.

- 1.) Go to the Service Desktop and select the **Diagnostics TAB** (See Figure 20).

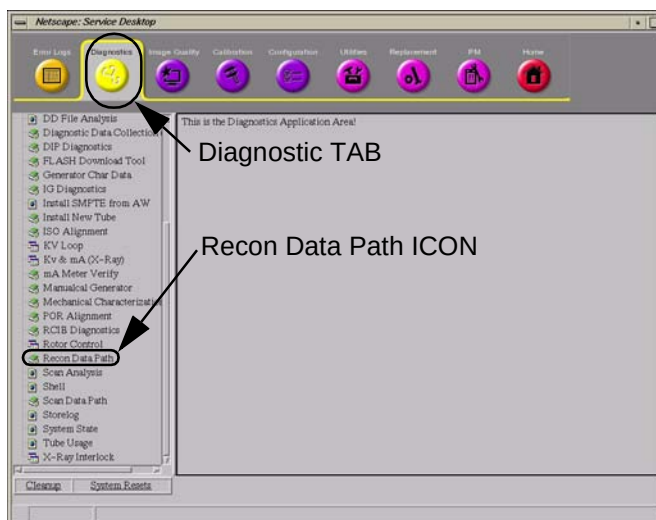


Figure 30 Diagnostic TAB Screen

- 2.) In the menu column on the left side of the Service Desktop page, click RECON DATA PATH icon (toward the bottom of the list). This brings up the Auto Recon page (takes 10 - 15 seconds to come up).

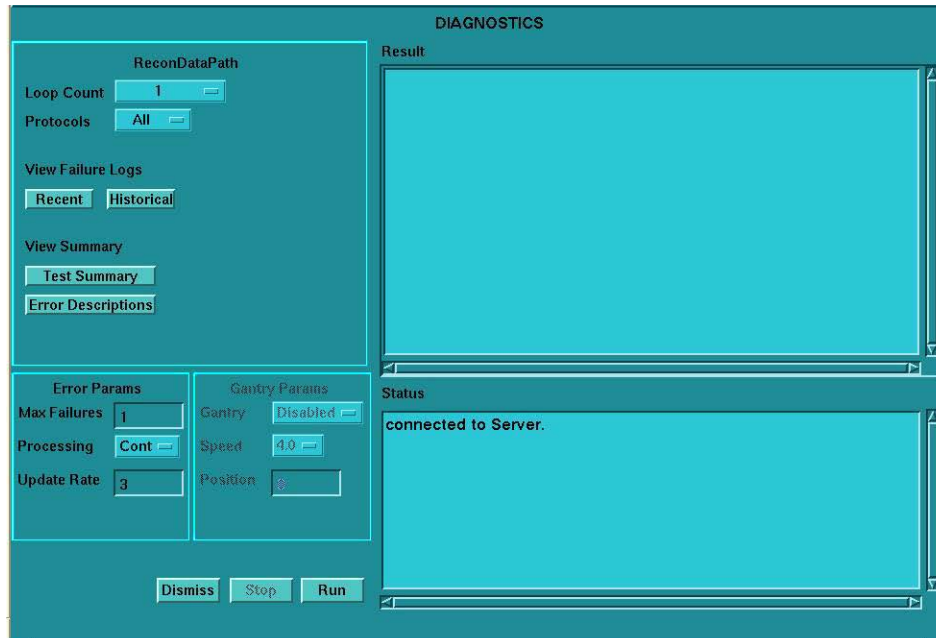


Figure 31 Recon Data Path Diagnostic Screen

- 3.) When the Recon Data Path page comes up, select a loop count of: 5, select ALL tests, then click RUN. This runs five loops of all tests; each loop generates 20 images, for a total of 100 images. This test checks the image output by comparing checksums of each image. (takes about 4 minutes to run).
- 4.) When complete, click DISMISS.

2 LED Status Information

The Diagnostic LED's can be visually inspected to assist in monitoring the various functions. They are visible with the console and ICE box covers removed.

LED OVERVIEW

DS17, DS18, DS19, and DS20 signify when the Xilinx FPGAs have completed their programming phase and are in application mode. (These should all go "on" about a half-second after power-up or board RESET).

DS12 and DS13 are user programmable via a register located in the UART serial port interface. (Currently not used during Diagnostics)

DS14, DS15, and DS16 indicate power supply status:

- DS14 - 5.0 Volt Supply is "up"
- DS15 - 2.6 Volt Supply is "up"
- DS16 - 1.9 Volt Supply is "up"

DS7-11 are user programmable via the FLAG(3) pin of the ADSP-21060 processor located in SDC-VW Processing Front-end. (These will blink during the "collision test" diagnostic).

DS1, DS2, DS3, DS4, and DS5 are user programmable via the FLAG(3) pin of the ADSP-21060 processor located in SDC-VW Processing Back-end. (These will blink during the "collision test" diagnostic).

DS11 is user programmable via the Timer-1 Out pin of the TMS320C6701 located in Filter Processing. (Currently not used during Diagnostics)

DS21-28 are user programmable via data bits 24-31 of the Post Processor.

Here are the functions of these LEDs during ROM-based diagnostics:

(Top) DS31 - (Unused)

DS30 - (Unused)

DS29 - APU LED. Blinks when the APU diags are running.

DS28 - C67 LED. Blinks when the C67 diags are running.

DS27 - IMAX LED. Blinks when the IMAX diags are running.

DS26 - SPAM1 LED. Blinks when the SPAM1 diags are running.

DS25 - SPAM0 LED. Blinks when the SPAM0 diags are running.

(Bottom) DS24 - VxWorks Heartbeat LED. Blinks when VxWorks is running.

During System tests, DS24 is the heartbeat LED for VxWorks.

DS29-36 are user programmable via data bits 24-31 of the Sigma_B Processor.

Here are the functions of these LEDs during ROM-based diagnostics:

(Top) DS31 - (Unused)

DS30 - (Unused)

DS29 - (Unused)

DS28 - (Unused)

DS27 - (Unused)

DS26 - . (Unused)

DS25 - Dual Port LED. Blinks when the Dual Port Diag is running.

(Bottom) DS24 - VxWorks Heartbeat LED. Blinks when VxWorks is running.

During System tests, DS24 is the heartbeat LED for VxWorks.

DS39 is active whenever the Image Memory's DMA Controller is active. (Blinks during IMAX and APU diagnostics)

DS38 is active whenever PCI data transfers are taking place.

DS37 is active whenever VME data transfers are taking place.

3 Interface Requirements

The Pegasus assembly interfaces to the outside world through the VME interface, the RS-232 57.6 Kbaud serial port located at J9, and the JTAG interfaces. Additionally, the Pegasus assembly has a PMC daughter card interface for future expansion.

4 Switches, Controls, and Adjustments

No controls or adjustments are part of this assembly.

A push button reset switch is provided on the faceplate for a manual hard reset of the board. Once this switch is released, a 300 millisecond active low reset signal is activated. This reset is equivalent to a power up reset.

End of Document